

CHAPTER - 7

COMPUTER COMMUNICATION AND NETWORKS

BASIC OF NETWORKS

7.1 Networks and its Advantages

1. Enhanced Communication and Collaboration:

One of the most significant advantages of networks is their ability to facilitate communication and collaboration among users. In a networked environment, individuals can easily share information, exchange messages, and collaborate on projects regardless of their physical location. For instance, through email, instant messaging, and video conferencing tools, team members can communicate in real-time, share files, and work together on documents simultaneously. This level of connectivity improves productivity and fosters efficient teamwork, enabling businesses to operate more effectively and organizations to achieve their goals more swiftly.



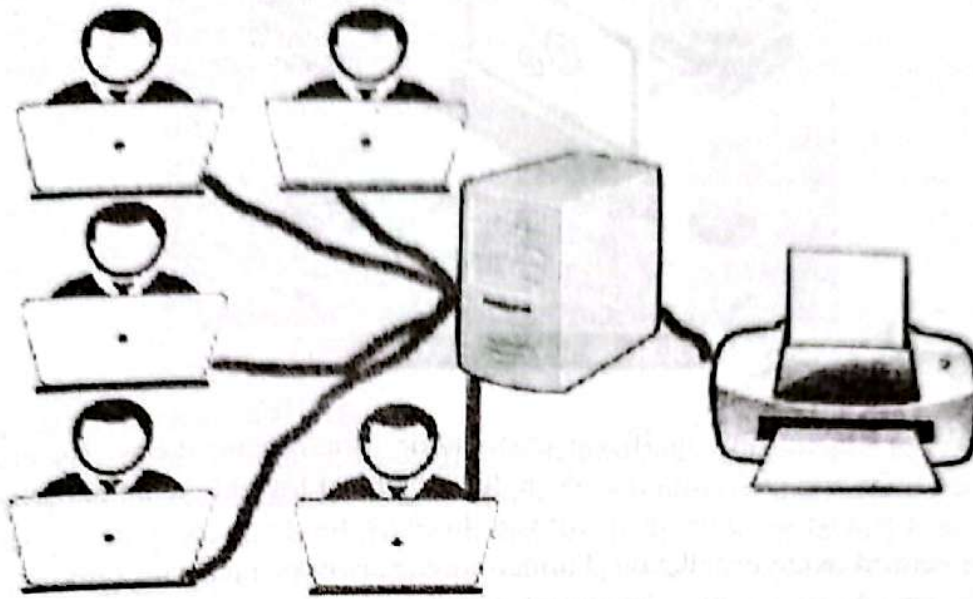
7.1 Communication and Collaboration

2. Resource Sharing:

Networks allow for the sharing of resources such as printers, scanners, and storage devices. In a networked environment, multiple users can access and utilize these shared resources, which reduces the need for duplicate equipment and lowers costs. For example, a single printer connected to a network can be used by all employees in an office, eliminating the need for multiple printers. Similarly, shared

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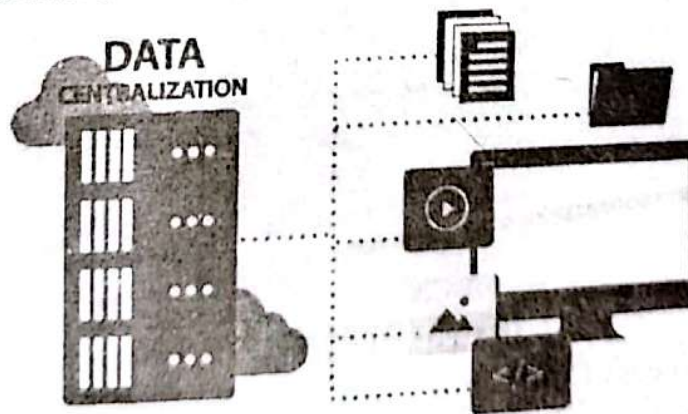
network storage solutions allow for centralized data management and backup, ensuring that important files are easily accessible and securely stored.



7.2 Resource Sharing

3. Centralized Data Management:

With networks, data can be centralized, meaning that information is stored in a central location rather than on individual devices. This centralization simplifies data management and enhances data security. Centralized data management allows for easier backups, more effective data recovery, and better control over data access. For instance, a company can maintain a central database that holds all its operational information, which can be accessed by authorized users from various locations. This setup not only streamlines data access but also ensures consistency and reduces the risk of data loss.



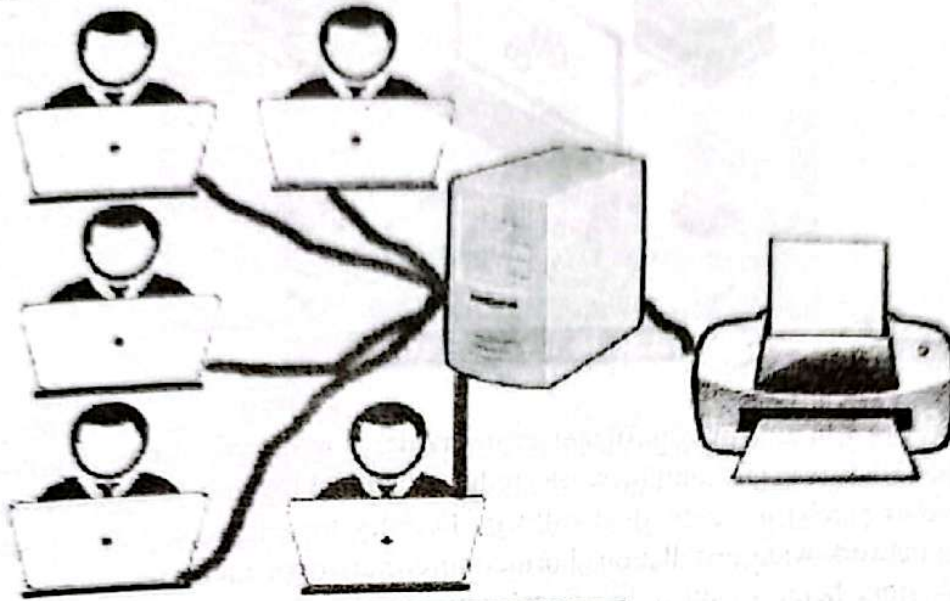
7.3 Centralized Data Management

4. Scalability and Flexibility:

Networks offer scalability and flexibility, allowing organizations to expand their infrastructure as needed. As a business grows, its network can be easily scaled to accommodate additional users, devices, and resources. This scalability is achieved by adding more network nodes, increasing bandwidth, or upgrading network components. Additionally, networks provide the flexibility to integrate new technologies and applications, enabling organizations to adapt to changing business needs and technological advancements. This adaptability ensures that the network can evolve alongside the organization's growth and development.

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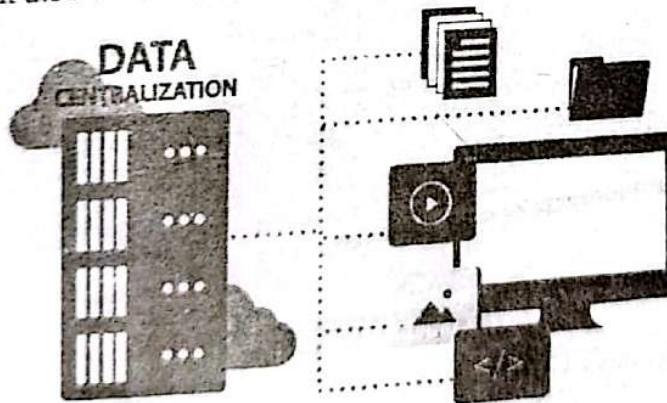
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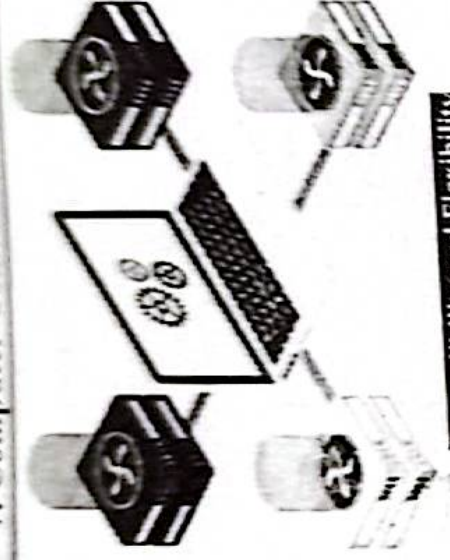
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7.4 Scalability and Flexibility

5. Cost Efficiency:

Implementing a network can lead to significant cost savings for organizations. By enabling resource sharing and centralized data management, networks reduce the need for redundant hardware and software. For example, instead of purchasing individual software licenses for each computer, a network can use a single license for a network-wide installation. Furthermore, networks facilitate remote access, allowing employees to work from home or other locations, which can reduce overhead costs associated with physical office spaces. The ability to centralize and manage resources effectively contributes to overall cost efficiency and budget optimization.



7.5 Cost Efficiency

6. Enhanced Security and Access Control:

Networks provide enhanced security features and access control mechanisms that help protect sensitive information. With network security measures such as firewalls, encryption, and access control lists, organizations can safeguard their data from unauthorized access and cyber threats. Network administrators can set up user permissions and access controls to ensure that only authorized individuals can access specific data and resources. Additionally, centralized security management allows for consistent application of security policies across the network, reducing the risk of security breaches and ensuring compliance with regulatory requirements.

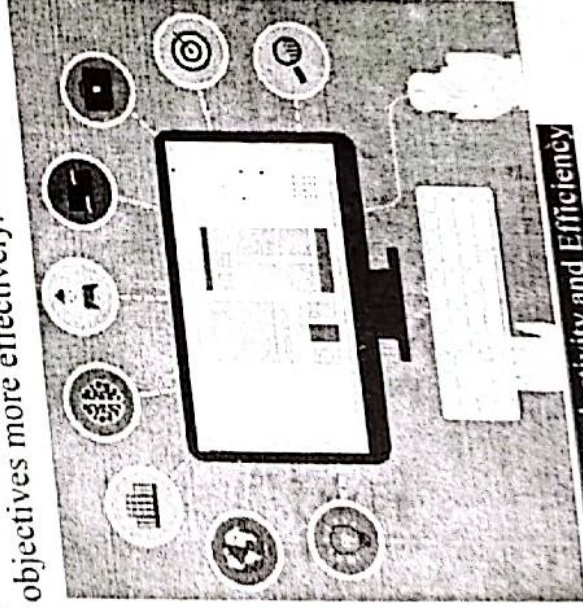
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7.6 Security and Access Control

7. Improved Productivity and Efficiency:

By facilitating communication, resource sharing, and centralized management, networks contribute to improved productivity and efficiency in the workplace. Employees can quickly access and share information, collaborate on projects, and utilize shared resources, leading to faster completion of tasks and more streamlined operations. The ability to automate routine processes and integrate various applications into a network also enhances operational efficiency, allowing organizations to focus on their core activities and achieve their objectives more effectively.



7.7 Productivity and Efficiency

In summary, networks play a crucial role in modern computing environments by enhancing communication, enabling resource sharing, centralizing data management, providing scalability and flexibility, reducing costs, improving security, and boosting productivity. These advantages make networks an essential component of any organization, supporting efficient operations and facilitating growth and innovation.

7.2 Networks Components

1. Sender

The sender is the originator of data or information in a communication process. This entity can be any device or application responsible for generating and transmitting a message across a network. Key points about the sender include:

Communication process by creating a message or data packet.

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- **Bandwidth:** Full-duplex systems typically require more bandwidth than simplex or half-duplex systems due to the need for simultaneous transmission and reception channels.
 - **Collision Detection:** In full-duplex communication, collision detection is not a concern as both directions are managed simultaneously. In half-duplex systems, mechanisms are needed to handle potential collisions when switching between transmission and reception.
- In summary, understanding the different data flow directions (simplex, half-duplex, and full-duplex) helps in selecting the appropriate communication mode for specific applications, balancing factors such as efficiency, complexity, and cost.

TRANSMISSION MEDIUM

7.4 Transmission Mediums and Their Types

1. Guided Medium

Guided mediums are physical transmission pathways that direct the flow of data signals from the sender to the receiver. Unlike unguided mediums, such as wireless communication, guided mediums confine the data signals to a specific path, helping to reduce interference and signal degradation. Here's a detailed look at the three primary types of guided mediums: twisted pair cables, coaxial cables, and fiber-optic cables.

1. Twisted Pair Cables:

Definition: Twisted pair cables consist of pairs of insulated copper wires twisted together. The twisting of wires helps to reduce electromagnetic interference and crosstalk from adjacent pairs and external sources.

Types:

- **Unshielded Twisted Pair (UTP):** Commonly used in Ethernet networks and telephone systems. UTP cables are cost-effective and suitable for a variety of applications, but they are not susceptible to electromagnetic interference compared to shielded options.
- **Shielded Twisted Pair (STP):** Includes additional shielding around the wires to provide better protection against interference. STP is used in environments with higher electromagnetic interference, offering improved performance over UTP.

Characteristics:

- **Construction:** Consists of pairs of wires twisted together to improve performance and reduce interference.
- **Bandwidth:** UTP cables typically support bandwidths up to 1 Gbps for Ethernet connections, while STP can support higher speeds and better performance in noisy environments.
- **Distance:** Effective for short to medium distances, usually up to 100 meters for standard Ethernet applications.

Applications: Used in local area networks (LANs), telephone lines, and some networking applications where cost and installation simplicity are key factors.

Advantages:

- **Cost-Effective:** Generally less expensive compared to coaxial and fiber-optic cables.
- **Ease of Installation:** Flexible and easy to install, making it suitable for various environments.

Limitations:

- **Interference:** UTP is more susceptible to electromagnetic interference (EMI) and crosstalk compared to STP and other guided mediums.

- ...: Versatile and widely used for various communication applications, including mobile communications. They offer broad coverage but can be susceptible to interference and limited bandwidth.
- **Microwaves:** Provide high bandwidth and are used for point-to-point and satellite communications. They require line-of-sight and can be affected by weather conditions.
 - **Infrared:** Used for short-range, line-of-sight communication, such as in remote controls and short-range data transfer. It offers good security but is limited by its range and sensitivity to obstacles.
- Each unguided medium has unique characteristics and is suited for different applications, balancing factors such as range, bandwidth, and susceptibility to interference.

NETWORK TYPES AND TOPOLOGIES

7.5 Types of Networks:

1. Local Area Networks (LAN)

A Local Area Network (LAN) is a network that connects computers and other devices within a limited geographical area, such as a home, office, or campus. LANs are designed to enable devices to share resources and communicate with each other efficiently. Here's a detailed look at LANs, their components, and their advantages:

1. Definition and Characteristics:

Definition: A LAN is a network that allows multiple devices to connect and communicate within a confined area, such as a single building or a group of nearby buildings. LANs are typically used for sharing resources like files, printers, and internet connections.

Characteristics:

- **Geographical Scope:** Limited to a small geographic area, typically ranging from a single room to several buildings.
- **High Data Transfer Rates:** Provides high-speed data transmission compared to wide area networks (WANs), often ranging from 10 Mbps to 10 Gbps or more, depending on the technology.

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- **Practicality:** The Internet Model is more aligned with the actual implementation of network protocols, making it more practical for real-world use, especially in comparison to the OSI Model, which is more of a theoretical framework.

Summary:

The Internet Model, or TCP/IP Model, is a four-layer framework that standardizes the processes and protocols involved in network communication. It consists of the Link Layer, Internet Layer, Transport Layer, and Application Layer, each responsible for specific functions essential for data transmission across networks. This model is the foundation of the internet and most modern networks, providing a practical, scalable, and widely adopted approach to networking. Its simplicity, flexibility, and interoperability make it an essential component of global communication technology.

7.2 NETWORK SECURITY

Common Types of Network Attacks

Network attacks are attempts to disrupt, disable, or gain unauthorized access to computer networks or networked systems. Here are some common types of network attacks:

1. Denial of Service (DoS) and Distributed Denial of Service (DDoS) Attacks

- **DoS:** An attacker floods a network or server with excessive traffic, overwhelming it to the point where legitimate users cannot access the services.
- **DDoS:** Similar to a DoS attack, but the traffic comes from multiple sources, often using a botnet, making it harder to defend against.

2. Man-in-the-Middle (MitM) Attack:

An attacker intercepts and possibly alters communication between two parties without their knowledge. This can lead to data theft, eavesdropping, or injecting malicious content into the communication.

3. Phishing and Spear Phishing:

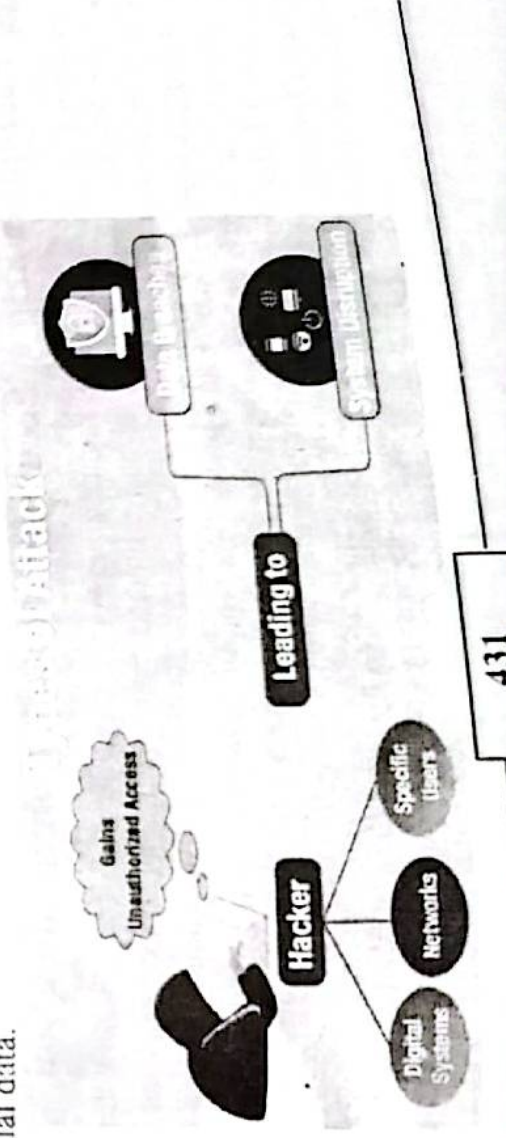
- **Phishing:** Attackers use deceptive emails or websites to trick users into providing sensitive information, such as passwords or credit card numbers.
- **Spear Phishing:** A more targeted form of phishing where the attacker tailors the attack to a specific individual or organization, often using personal information to appear legitimate.

4. SQL Injection:

Attackers exploit vulnerabilities in web applications by injecting malicious SQL code into the application's input fields. This can lead to unauthorized access to the database, allowing attackers to read, modify, or delete data.

5. Packet Sniffing (Eavesdropping):

Attackers use packet-sniffing tools to capture and analyze data packets traveling over a network. This can expose sensitive information such as passwords, usernames, and other confidential data.



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6. Rogue Access Points: Attackers set up unauthorized wireless access points to lure users into connecting to them. Once connected, the attacker can monitor the traffic and potentially gain access to the user's device.

7. Password Attacks:

- **Brute Force:** Attackers try all possible password combinations until they find the correct one.
- **Dictionary Attack:** Attackers use a list of common passwords or words to try and gain access.
- **Credential Stuffing:** Attackers use credentials obtained from data breaches to attempt access on other platforms, banking on users reusing passwords.

8. Ransomware: Malicious software encrypts the victim's data, rendering it inaccessible. The attacker demands a ransom payment in exchange for the decryption key.

9. ARP Spoofing: Attackers send falsified ARP (Address Resolution Protocol) messages to a local network. This can result in the attacker intercepting, modifying, or blocking data intended for another device on the network.

10. DNS Spoofing: Also known as DNS cache poisoning, attackers corrupt the DNS server's cache to redirect traffic from a legitimate site to a fraudulent one. This is often used in conjunction with phishing attacks.

11. Cross-Site Scripting (XSS): Attackers inject malicious scripts into web pages viewed by other users. These scripts can steal cookies, session tokens, or other sensitive information.

12. Malware Injection: Malicious software (malware) is introduced into a network, often through email attachments, infected websites, or compromised software. Once inside, the malware can spread, disrupt operations, or steal data.

13. Session Hijacking: Attackers exploit an active session between a client and a server to gain unauthorized access. By stealing the session token, the attacker can impersonate the user and perform unauthorized actions.

14. Zero-Day Exploits: These attacks exploit previously unknown vulnerabilities in software or hardware. Since the vulnerability is not yet patched, these attacks can be particularly damaging.

15. Insider Threats: Malicious insiders, such as disgruntled employees, may use their access to a network to steal data, disrupt operations, or facilitate further attacks.

16. Advanced Persistent Threats (APTs): APTs are prolonged and targeted attacks where the attacker infiltrates a network and remains undetected for an extended period, gathering intelligence or stealing data.

Understanding these types of attacks is crucial for implementing effective security measures and protecting networks from potential threats.

7.9 Encryption help in Securing Network Communication

Encryption plays a crucial role in securing network communication by ensuring that data transmitted between devices is kept confidential and protected from unauthorized access. Here's how encryption helps secure network communication:

1. **Confidentiality: Data Encryption:** Encryption converts plain text data into a coded form (ciphertext) that is unreadable without the proper decryption key. This ensures that even if the data is intercepted by unauthorized parties, it remains unusable.